

A Practical Guide to AlphaFold

TOPIC -- ALPHAFOLD

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Derived databases AlphaFill adds cofactors to AlphaFold models where appropriate. This is achieved by searching the Protein Data Bank for similar structures and transplanting cofactors to analogous positions. It is also linked to by UniProt. TmAlphaFold docks AlphaFold models to biological membranes, similar to what OPM does for PDB structures. AFTM uses AlphaFold models to identify transmembrane regions in human proteins, similar to what PDBTM does for PDB structures. ChannelsDB 2.0 uses PDB or AlphaFill models to calculate the pathway a molecule may take to reach an enzyme's active site or to reach another side of a transmembrane pore. The AFDB is not updated with UniProt sequences changes. AlphaSync keeps the AFDB in sync with UniProt entry changes, generating updated structures, residue-level features and contacts. It tries to use an AFDB entry for the exact updated sequence when available and run AlphaFold 2 independently otherwise. It fills in AFDB's blank for large (> 2,700 aa) proteins and proteins with special FASTA characters such as B, Z, U or X. The Encyclopedia of Domains (TED) applies the domain-recognition method from CATH database to 188 million unique structures from the AFDB, identifying nearly 365 million domains, which is 100 million more than what sequence-based methods could identify. The Evolutionary Classification of Protein Domains database (ECOD) assigns ECOD classifications to all SwissProt proteins in AFDB.

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Further reading Carlos Outeiral, CASP14: what Google DeepMind's AlphaFold 2 really achieved, and what it means for protein folding, biology and bioinformatics, Oxford Protein Informatics Group. (3 December) Mohammed AlQuraishi, AlphaFold2 @ CASP14: "It feels like one's child has left home." (blog), 8 December 2020 Mohammed AlQuraishi, The AlphaFold2 Method Paper: A Fount of Good Ideas (blog), 25 July 2021

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CASP13 In December 2018, DeepMind's AlphaFold placed first in the overall rankings of the 13th Critical Assessment of Techniques for Protein Structure Prediction (CASP). The program was particularly successfully predicting the most accurate structure for targets rated as the most difficult by the competition organisers, where no existing template structures were available from proteins

with a partially similar sequence. AlphaFold gave the best prediction for 25 out of 43 protein targets in this class, achieving a median score of 58.9 on the CASP's global distance test (GDT) score, ahead of 52.5 and 52.4 by the two next best-placed teams, who were also using deep learning to estimate contact distances. Overall, across all targets, AlphaFold 1 achieved a GDT score of 68.5. In January 2020, implementations and illustrative code of AlphaFold 1 was released open-source on GitHub. but, as stated in the "Read Me" file on that website: "This code can't be used to predict structure of an arbitrary protein sequence. It can be used to predict structure only on the CASP13 dataset (links below). The feature generation code is tightly coupled to our internal infrastructure as well as external tools, hence we are unable to open-source it." Therefore, in essence, the code deposited is not suitable for general use but only for the CASP13 proteins. The company has not announced plans to make their code publicly available as of 5 March 2021.

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AlphaFold 1 (2018) was built on work developed by various teams in the 2010s, work that looked at the large databases of related protein sequences now available from many different organisms (most without known 3D structures), to try to find changes at different residues (peptides) that appeared to be correlated, even though the residues were not consecutive in the main chain. Such correlations suggest that the residues may be close to each other physically, even though not close in the sequence, allowing a contact map to be estimated. Building on recent work prior to 2018, AlphaFold 1 extended this by estimating a probability distribution for the distances between residues, effectively transforming the contact map into a distance map. It also used more advanced learning methods than previously to develop the inference. The code was not made publicly available, except to run on sequences of proteins in the 2018 CASP competition.

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Published works Andrew W. Senior et al. (December 2019), "Protein structure prediction using multiple deep neural networks in the 13th Critical Assessment of Protein Structure Prediction (CASP13)", *Proteins: Structure, Function, Bioinformatics* 87(12) 1141–1148 doi:10.1002/prot.25834 Andrew W. Senior et al. (15 January 2020), "Improved protein structure prediction using potentials from deep learning", *Nature* 577 706–710 doi:10.1038/s41586-019-1923-7 John Jumper et al. (December 2020), "High Accuracy Protein Structure Prediction Using Deep Learning", in *Fourteenth Critical Assessment of Techniques for Protein Structure Prediction (Abstract Book)*, pp. 22–24 John Jumper et al.

(December 2020), "AlphaFold 2". Presentation given at CASP 14. Abramson, J., Adler, J., Dunger, J. et al. (May 2024), "Accurate structure prediction of biomolecular interactions with AlphaFold 3", Nature 630, 493–500 (2024)

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The 2020 version of the program (AlphaFold 2, 2020) is significantly different from the original version that won CASP 13 in 2018, according to the team at DeepMind. AlphaFold 1 used a number of separately trained modules to produce a guide potential, which was then combined with a physics-based energy potential. AlphaFold 2 replaced this with a system of interconnected sub-networks, forming a single, differentiable, end-to-end model based on pattern recognition. This model was trained in an integrated manner. After the neural network's prediction converges, a final refinement step applies local physical constraints using energy minimization based on the AMBER force field. This step only slightly adjusts the predicted structure. A key part of the 2020 system are two modules, believed to be based on a transformer design, which are used to progressively refine a vector of information for each relationship (or "edge" in graph-theory terminology) between an amino acid residue of the protein and another amino acid residue (these relationships are represented by the array shown in green); and between each amino acid position and each different sequences in the input sequence alignment (these relationships are represented by the array shown in red). Internally these refinement transformations contain layers that have the effect of bringing relevant data together and filtering out irrelevant data (the "attention mechanism") for these relationships, in a context-dependent way, learned from training data. These transformations are iterated, the updated information output by one step becoming the input of the next, with the sharpened residue/residue information feeding into the update of the residue/sequence information, and then the improved residue/sequence information feeding into the update of the residue/residue information. As the iteration progresses, according to one report, the "attention algorithm ... mimics the way a person might assemble a jigsaw puzzle: first connecting pieces in small clumps—in this case clusters of amino acids—and then searching for ways to join the clumps in a larger whole." The output of these iterations then informs the final structure prediction module, which also uses transformers, and is itself then iterated. In an example presented by DeepMind, the structure prediction module achieved a correct topology for the target protein on its first iteration, scored as having a GDT_TS of 78, but with a large number (90%) of stereochemical violations – i.e. unphysical bond angles or lengths. With subsequent iterations the number of stereochemical violations fell. By the third iteration the GDT_TS of the prediction was approaching 90, and by the eighth iteration the number of stereochemical

violations was approaching zero. The training data was originally restricted to single peptide chains. However, the October 2021 update, named AlphaFold-Multimer, included protein complexes in its training data. DeepMind stated this update succeeded about 70% of the time at accurately predicting protein-protein interactions.